PGCP-AC AUGUST 26
Assignment No- 2

1. Write a SQL query to create Employee & Department table with specific columns and constraints.(Note- Create Table, insert data as per your Choice)

```
mysql> CREATE DATABASE CompanyDB;
Query OK, 1 row affected (0.01 sec)

mysql> USE CompanyDB;
Database changed
mysql> CREATE TABLE Emp()
   -> ^C
mysql> CREATE TABLE Emp(
   -> EmpID INT PRIMARY KEY
   -> ^C
mysql> CREATE TABLE Emp(
   -> EmpID INT PRIMARY KEY,
   -> Name VARCHAR(100) NOT NULL,
   -> Salary DECIMAL(10,2),
   -> DeptID INT,
   -> JoiningDate DATE,
   -> JobTitle VARCHAR(50),
   -> City VARCHAR(50));
Query OK, 0 rows affected (0.02 sec)

mysql> CREATE TABLE Department(
   -> DeptID INT PRIMARY KEY,
   -> DeptName VARCHAR(100) NOT NULL);
Query OK, 0 rows affected (0.01 sec)
```

```
mysql> SHOW TABLES;
+----------------------+
| Tables_in_companydb |
+----------------------+
| department           |
| emp                  |
+----------------------+
2 rows in set (0.00 sec)

mysql> desc Emp;
+-------------+---------------+------+-----+---------+-------+
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| EmpID       | int           | NO   | PRI | NULL    |       |
| Name        | varchar(100)  | NO   |     | NULL    |       |
| Salary      | decimal(10,2) | YES  |     | NULL    |       |
| DeptID      | int           | YES  |     | NULL    |       |
| JoiningDate | date          | YES  |     | NULL    |       |
| JobTitle    | varchar(50)   | YES  |     | NULL    |       |
| City        | varchar(50)   | YES  |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
7 rows in set (0.00 sec)

mysql> DESC Department;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| DeptID   | int          | NO   | PRI | NULL    |       |
| DeptName | varchar(100) | NO   |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
2 rows in set (0.00 sec)
```

2. Write a SQL query to add any new column to an existing table.

```
mysql> ALTER TABLE Department
   -> ADD COLUMN Salary DECIMAL(10,2);
Query OK, 0 rows affected (0.01 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

```
mysql> DESC Department;
+----------+--------------+------+-----+---------+-------+
| Field    | Type         | Null | Key | Default | Extra |
+----------+--------------+------+-----+---------+-------+
| DeptID   | int          | NO   | PRI | NULL    |       |
| DeptName | varchar(100) | NO   |     | NULL    |       |
| Salary   | decimal(10,2)| YES  |     | NULL    |       |
+----------+--------------+------+-----+---------+-------+
3 rows in set (0.00 sec)
```

3. Write a SQL query to insert multiple records (At least 5) into a table in a single operation.

```
mysql> INSERT INTO Emp (EmpID, Name, Salary, DeptID, JoiningDate, JobTitle, City)
   -> VALUES
   -> (101, 'Rahul', 45000, 10, '2024-01-15', 'Developer', 'Mumbai'),
   -> (102, 'Amit', 55000, 20, '2023-06-10', 'Manager', 'Pune'),
   -> (103, 'Priya', 60000, 10, '2022-03-20', 'Salesperson', 'Mumbai'),
   -> (104, 'Sneha', 40000, 30, '2024-07-01', 'Tester', 'Delhi'),
   -> (105, 'Vijay', 52000, 20, '2023-11-12', 'Salesperson', 'Mumbai');
Query OK, 5 rows affected (0.01 sec)
Records: 5  Duplicates: 0  Warnings: 0
```

```
mysql> SELECT * FROM Emp;
+-------+-------+----------+--------+-------------+-------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle    | City   |
+-------+-------+----------+--------+-------------+-------------+--------+
|   101 | Rahul | 45000.00 |     10 | 2024-01-15  | Developer   | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager     | Pune   |
|   103 | Priya | 60000.00 |     10 | 2022-03-20  | Salesperson | Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester      | Delhi  |
|   105 | Vijay | 52000.00 |     20 | 2023-11-12  | Salesperson | Mumbai |
+-------+-------+----------+--------+-------------+-------------+--------+
5 rows in set (0.00 sec)
```

4. Write a SQL query to delete all records from a table.

```
mysql> DELETE FROM Emp;
Query OK, 5 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM Emp;
Empty set (0.00 sec)
```

5. Write a SQL query to update salary records from a table where id = 5.

```
mysql> UPDATE Emp
   -> SET Salary=49999
   -> WHERE EmpID = 105;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

```
mysql> SELECT * FROM Emp;
+-------+-------+----------+--------+-------------+-------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle    | City   |
+-------+-------+----------+--------+-------------+-------------+--------+
|   101 | Rahul | 45000.00 |     10 | 2024-01-15  | Developer   | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager     | Pune   |
|   103 | Priya | 60000.00 |     10 | 2022-03-20  | Salesperson | Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester      | Delhi  |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson | Mumbai |
+-------+-------+----------+--------+-------------+-------------+--------+
5 rows in set (0.00 sec)
```

6. Write a query to grant SELECT privilege on the Emp table to user readonly_user.

```
mysql> CREATE USER 'readonly_user'@'localhost'
   -> IDENTIFIED BY 'Test@123';
Query OK, 0 rows affected (0.02 sec)

mysql> GRANT SELECT ON CompanyDB.Emp
   -> TO 'readonly_user'@'localhost';
Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SHOW GRANTS FOR 'readonly_user'@'localhost';
+-----------------------------------------------------------------+
| Grants for readonly_user@localhost                              |
+-----------------------------------------------------------------+
| GRANT USAGE ON *.* TO `readonly_user`@`localhost`               |
| GRANT SELECT ON `companydb`.`emp` TO `readonly_user`@`localhost` |
+-----------------------------------------------------------------+
2 rows in set (0.00 sec)
```

7. Write a query to revoke INSERT privilege on the Emp table from user temp_user.

```
mysql> CREATE USER 'temp_user'@'localhost'
    -> IDENTIFIED BY 'Temp@123';
Query OK, 0 rows affected (0.00 sec)

mysql> GRANT INSERT ON CompanyDB.Emp
    -> TO 'temp_user'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> SHOW GRANTS FOR 'temp_user'@'localhost';
+-----------------------------------------------------------------+
| Grants for temp_user@localhost                                  |
+-----------------------------------------------------------------+
| GRANT USAGE ON *.* TO `temp_user`@`localhost`                   |
| GRANT INSERT ON `companydb`.`emp` TO `temp_user`@`localhost`    |
+-----------------------------------------------------------------+
2 rows in set (0.00 sec)

mysql> REVOKE INSERT ON CompanyDB.Emp
    -> FROM 'temp_user'@'localhost';
Query OK, 0 rows affected (0.00 sec)

mysql> SHOW GRANTS FOR 'temp_user'@'localhost';
+-------------------------------------------------+
| Grants for temp_user@localhost                  |
+-------------------------------------------------+
| GRANT USAGE ON *.* TO `temp_user`@`localhost`   |
+-------------------------------------------------+
1 row in set (0.00 sec)
```

8. Write a query to insert a new employee into the Emp table and commit the transaction.

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> INSERT INTO Emp
   -> (EmpID, Name, Salary, DeptID, JoiningDate, JobTitle, City)
   -> VALUES
   -> (106, 'Karan', 48000, 10, '2025-01-10', 'Developer', 'Mumbai');
Query OK, 1 row affected (0.00 sec)

mysql>
mysql> COMMIT;

Query OK, 0 rows affected (0.00 sec)
```

```
mysql> SELECT * FROM Emp;
+-------+-------+----------+--------+-------------+------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle   | City   |
+-------+-------+----------+--------+-------------+------------+--------+
|   101 | Rahul | 59999.00 |     10 | 2024-01-15  | Developer  | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager    | Pune   |
|   103 | Priya | 60000.00 |     10 | 2022-03-20  | Salesperson| Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester     | Delhi  |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson| Mumbai |
|   106 | Karan | 48000.00 |     10 | 2025-01-10  | Developer  | Mumbai |
+-------+-------+----------+--------+-------------+------------+--------+
6 rows in set (0.00 sec)
```

9. Write a query to update the Salary of an employee with EmpID = 101 and commit the transaction.

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE Emp
   -> SET Salary= 59999
   -> WHERE EmpID = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql> SELECT EmpID, Name, Salary
   -> FROM Emp
   -> WHERE EmpID = 101;
+-------+-------+----------+
| EmpID | Name  | Salary   |
+-------+-------+----------+
|   101 | Rahul | 59999.00 |
+-------+-------+----------+
1 row in set (0.00 sec)
```

10. Write a query to set a SAVEPOINT before updating the DeptID of an employee.

```
+-------+-------+----------+--------+-------------+------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle   | City   |
+-------+-------+----------+--------+-------------+------------+--------+
|   101 | Rahul | 59999.00 |     10 | 2024-01-15  | Developer  | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager    | Pune   |
|   103 | Priya | 60000.00 |     10 | 2022-03-20  | Salesperson| Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester     | Delhi  |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson| Mumbai |
|   106 | Karan | 48000.00 |     10 | 2025-01-10  | Developer  | Mumbai |
+-------+-------+----------+--------+-------------+------------+--------+
```

```
mysql> START TRANSACTION;
Query OK, 0 rows affected (0.00 sec)

mysql> SAVEPOINT sp1;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE Emp
    -> SET DeptID = 20
    -> WHERE EmpID = 101;
Query OK, 1 row affected (0.00 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql> SELECT * FROM Emp;
+-------+-------+---------+--------+-------------+-------------+--------+
| EmpID | Name  | Salary  | DeptID | JoiningDate | JobTitle    | City   |
+-------+-------+---------+--------+-------------+-------------+--------+
|   101 | Rahul | 59999.00 |     20 | 2024-01-15  | Developer   | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager     | Pune   |
|   103 | Priya | 60000.00 |     10 | 2022-03-20  | Salesperson | Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester      | Delhi  |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson | Mumbai |
|   106 | Karan | 48000.00 |     10 | 2025-01-10  | Developer   | Mumbai |
+-------+-------+---------+--------+-------------+-------------+--------+
6 rows in set (0.00 sec)
```

11. Write a query to rollback to the previously created SAVEPOINT.

mysql> ROLLBACK TO SAVEPOINT sp1;

Query OK, 0 rows affected (0.00 sec)

```
mysql> SELECT * FROM Emp;
+-------+-------+---------+--------+-------------+-------------+--------+
| EmpID | Name  | Salary  | DeptID | JoiningDate | JobTitle    | City   |
+-------+-------+---------+--------+-------------+-------------+--------+
|   101 | Rahul | 59999.00 |     10 | 2024-01-15  | Developer   | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager     | Pune   |
|   103 | Priya | 60000.00 |     10 | 2022-03-20  | Salesperson | Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester      | Delhi  |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson | Mumbai |
|   106 | Karan | 48000.00 |     10 | 2025-01-10  | Developer   | Mumbai |
+-------+-------+---------+--------+-------------+-------------+--------+
6 rows in set (0.00 sec)
```

12. Write a query to truncate all data from the Emp table.

I CREATE TEMP_EMP AND I GOING TO DELETE TEMP_EMP

```
mysql> CREATE TABLE Emp_Temp LIKE Emp;
Query OK, 0 rows affected (0.02 sec)

mysql> INSERT INTO EMP_TEMP
    -> SELECT * FROM Emp;
Query OK, 6 rows affected (0.01 sec)
Records: 6  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Emp;
+-------+-------+----------+--------+-------------+------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle   | City   |
+-------+-------+----------+--------+-------------+------------+--------+
|   101 | Rahul | 59999.00 |     10 | 2024-01-15  | Developer  | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager    | Pune   |
|   103 | Priya | 60000.00 |     10 | 2022-03-20  | Salesperson| Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester     | Delhi  |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson| Mumbai |
|   106 | Karan | 48000.00 |     10 | 2025-01-10  | Developer  | Mumbai |
+-------+-------+----------+--------+-------------+------------+--------+
6 rows in set (0.00 sec)

mysql> SELECT * FROM Emp_Temp;
+-------+-------+----------+--------+-------------+------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle   | City   |
+-------+-------+----------+--------+-------------+------------+--------+
|   101 | Rahul | 59999.00 |     10 | 2024-01-15  | Developer  | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager    | Pune   |
|   103 | Priya | 60000.00 |     10 | 2022-03-20  | Salesperson| Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester     | Delhi  |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson| Mumbai |
|   106 | Karan | 48000.00 |     10 | 2025-01-10  | Developer  | Mumbai |
+-------+-------+----------+--------+-------------+------------+--------+
6 rows in set (0.00 sec)
```

```
mysql> TRUNCATE TABLE Emp_Temp;
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM Emp_Temp;
Empty set (0.00 sec)

mysql> DESC EMP_TEMP;
+-------------+---------------+------+-----+---------+-------+
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| EmpID       | int           | NO   | PRI | NULL    |       |
| Name        | varchar(100)  | NO   |     | NULL    |       |
| Salary      | decimal(10,2) | YES  |     | NULL    |       |
| DeptID      | int           | YES  |     | NULL    |       |
| JoiningDate | date          | YES  |     | NULL    |       |
| JobTitle    | varchar(50)   | YES  |     | NULL    |       |
| City        | varchar(50)   | YES  |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
7 rows in set (0.00 sec)
```

13. Write a query to list all employees whose salary is greater than 50,000.

```
mysql> SELECT * FROM Emp;
+-------+-------+----------+--------+-------------+-------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle    | City   |
+-------+-------+----------+--------+-------------+-------------+--------+
|   101 | Rahul | 59999.00 |     10 | 2024-01-15  | Developer   | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager     | Pune   |
|   103 | Priya | 60000.00 |     10 | 2022-03-20  | Salesperson | Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester      | Delhi  |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson | Mumbai |
|   106 | Karan | 48000.00 |     10 | 2025-01-10  | Developer   | Mumbai |
+-------+-------+----------+--------+-------------+-------------+--------+
6 rows in set (0.00 sec)

mysql> SELECT EmpID, Name, Salary
    -> FROM Emp
    -> WHERE Salary > 50000;
+-------+-------+----------+
| EmpID | Name  | Salary   |
+-------+-------+----------+
|   101 | Rahul | 59999.00 |
|   102 | Amit  | 55000.00 |
|   103 | Priya | 60000.00 |
+-------+-------+----------+
3 rows in set (0.00 sec)
```

14. Write a query to increase the salary of all employees in DeptID = 10 by 10%.

```
mysql> UPDATE Emp
    -> SET Salary = Salary + (Salary * 10 / 100)
    -> WHERE DeptID = 10;
Query OK, 3 rows affected (0.00 sec)
Rows matched: 3  Changed: 3  Warnings: 0

mysql> SELECT * FROM EMP;
+-------+-------+----------+--------+-------------+-------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle    | City   |
+-------+-------+----------+--------+-------------+-------------+--------+
|   101 | Rahul | 65998.90 |     10 | 2024-01-15  | Developer   | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager     | Pune   |
|   103 | Priya | 66000.00 |     10 | 2022-03-20  | Salesperson | Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester      | Delhi  |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson | Mumbai |
|   106 | Karan | 52800.00 |     10 | 2025-01-10  | Developer   | Mumbai |
+-------+-------+----------+--------+-------------+-------------+--------+
6 rows in set (0.00 sec)
```

15. Write a query to count the total number of employees in each department.

```
mysql> SELECT DeptID, COUNT(*) AS TotalEmployees
    -> FROM Emp
    -> GROUP BY DeptID;
+--------+----------------+
| DeptID | TotalEmployees |
+--------+----------------+
|     10 |              3 |
|     20 |              2 |
|     30 |              1 |
+--------+----------------+
3 rows in set (0.00 sec)
```

16. Write a query to list departments that have no employees. & create a view for those salespeople who belong to the city of Mumbai.

```
mysql> SELECT * FROM Department;
+--------+-----------+----------+
| DeptID | DeptName  | Salary   |
+--------+-----------+----------+
|     10 | IT        | 50000.00 |
|     20 | Sales     | 45000.00 |
|     30 | HR        | 40000.00 |
|     40 | Finance   | 35000.00 |
|     50 | Marketing | 42000.00 |
+--------+-----------+----------+
5 rows in set (0.00 sec)

mysql> SELECT * FROM Emp;
+-------+-------+----------+--------+-------------+------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle   | City   |
+-------+-------+----------+--------+-------------+------------+--------+
|   101 | Rahul | 65998.90 |     10 | 2024-01-15  | Developer  | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager    | Pune   |
|   103 | Priya | 66000.00 |     10 | 2022-03-20  | Salesperson| Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester     | Delhi  |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson| Mumbai |
|   106 | Karan | 52800.00 |     10 | 2025-01-10  | Developer  | Mumbai |
+-------+-------+----------+--------+-------------+------------+--------+
6 rows in set (0.00 sec)

mysql> SELECT d.DeptID , d.DeptName
    -> FROM Department d
    -> LEFT JOIN emp e
    -> ON d.DeptID = e.DeptID
    -> WHERE e.EmpID IS NULL;
+--------+-----------+
| DeptID | DeptName  |
+--------+-----------+
|     40 | Finance   |
|     50 | Marketing |
+--------+-----------+
2 rows in set (0.00 sec)
```

```
mysql> CREATE VIEW MumbaiSalesPeople AS
    -> SELECT EmpID, Name, Salary, City
    -> FROM Emp
    -> WHERE JobTitle = 'Salesperson'
    -> AND City = 'Mumbai';
Query OK, 0 rows affected (0.01 sec)

mysql> SELECT * FROM MumbaiSalesPeople;
+-------+-------+----------+--------+
| EmpID | Name  | Salary   | City   |
+-------+-------+----------+--------+
|   103 | Priya | 66000.00 | Mumbai |
|   105 | Vijay | 49999.00 | Mumbai |
+-------+-------+----------+--------+
2 rows in set (0.00 sec)
```

17. Write a query to display the highest salary among all employees.

```
mysql> SELECT Salary FROM Emp
    -> ORDER BY Salary DESC
    -> LIMIT 1;
+----------+
| Salary   |
+----------+
| 66000.00 |
+----------+
1 row in set (0.00 sec)

mysql> SELECT EmpID, Name, Salary
    -> FROM Emp
    -> ORDER BY Salary DESC
    -> LIMIT 1;
+-------+-------+----------+
| EmpID | Name  | Salary   |
+-------+-------+----------+
|   103 | Priya | 66000.00 |
+-------+-------+----------+
1 row in set (0.00 sec)

mysql> SELECT EmpID, Name, Salary
    -> FROM Emp
    -> WHERE Salary = (
    -> SELECT MAX(Salary)
    -> FROM Emp);
+-------+-------+----------+
| EmpID | Name  | Salary   |
+-------+-------+----------+
|   103 | Priya | 66000.00 |
+-------+-------+----------+
1 row in set (0.00 sec)

mysql> SELECT EmpID, Name, Salary
    -> FROM Emp
    -> WHERE Salary >= ALL (
    -> SELECT Salary
    -> FROM Emp
    -> WHERE Salary IS NOT NULL);
+-------+-------+----------+
| EmpID | Name  | Salary   |
+-------+-------+----------+
|   103 | Priya | 66000.00 |
+-------+-------+----------+
1 row in set (0.00 sec)
```

```
mysql> SELECT MAX(Salary) AS HighestSalary
    -> FROM Emp;
+---------------+
| HighestSalary |
+---------------+
|      66000.00 |
+---------------+
1 row in set (0.00 sec)
```

18. Write a query to display the 2nd highest salary among all employees.

```
mysql> SELECT MAX(Salary) AS SecondHighestSalary
    -> FROM Emp
    -> WHERE Salary<(
    -> SELECT MAX(Salary)
    -> FROM Emp);
+---------------------+
| SecondHighestSalary |
+---------------------+
|            65998.90 |
+---------------------+
1 row in set (0.00 sec)

mysql> SELECT Name, Salary
    -> FROM Emp
    -> WHERE Salary = (
    -> SELECT MAX(Salary)
    -> FROM Emp
    -> WHERE Salary < (
    -> SELECT MAX(Salary)
    -> FROM Emp));
+-------+----------+
| Name  | Salary   |
+-------+----------+
| Rahul | 65998.90 |
+-------+----------+
1 row in set (0.00 sec)

mysql> SELECT DISTINCT Salary FROM Emp
    -> ORDER BY Salary DESC
    -> LIMIT 1 OFFSET 1;
+----------+
| Salary   |
+----------+
| 65998.90 |
+----------+
1 row in set (0.00 sec)

mysql> SELECT DISTINCT Salary,Name FROM Emp
    -> ORDER BY Salary DESC
    -> LIMIT 1 OFFSET 1;
+----------+-------+
| Salary   | Name  |
+----------+-------+
| 65998.90 | Rahul |
+----------+-------+
1 row in set (0.00 sec)
```

19. Write a query to show all employees ordered by JoiningDate in descending order.

```
mysql> SELECT * FROM Emp;
+-------+-------+----------+--------+-------------+-------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle    | City   |
+-------+-------+----------+--------+-------------+-------------+--------+
|   101 | Rahul | 65998.90 |     10 | 2024-01-15  | Developer   | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager     | Pune   |
|   103 | Priya | 66000.00 |     10 | 2022-03-20  | Salesperson | Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester      | Delhi  |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson | Mumbai |
|   106 | Karan | 52800.00 |     10 | 2025-01-10  | Developer   | Mumbai |
+-------+-------+----------+--------+-------------+-------------+--------+
6 rows in set (0.00 sec)
```

```
mysql> SELECT * FROM Emp
    -> ORDER BY JoiningDate DESC;
+-------+-------+----------+--------+-------------+-------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle    | City   |
+-------+-------+----------+--------+-------------+-------------+--------+
|   106 | Karan | 52800.00 |     10 | 2025-01-10  | Developer   | Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester      | Delhi  |
|   101 | Rahul | 65998.90 |     10 | 2024-01-15  | Developer   | Mumbai |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager     | Pune   |
|   103 | Priya | 66000.00 |     10 | 2022-03-20  | Salesperson | Mumbai |
+-------+-------+----------+--------+-------------+-------------+--------+
6 rows in set (0.00 sec)
```

20. Write a query to rename the DeptName column to DepartmentName.

```
mysql> SELECT * FROM Emp
    -> ORDER BY JoiningDate DESC;
+-------+-------+----------+--------+-------------+-------------+--------+
| EmpID | Name  | Salary   | DeptID | JoiningDate | JobTitle    | City   |
+-------+-------+----------+--------+-------------+-------------+--------+
|   106 | Karan | 52800.00 |     10 | 2025-01-10  | Developer   | Mumbai |
|   104 | Sneha | 40000.00 |     30 | 2024-07-01  | Tester      | Delhi  |
|   101 | Rahul | 65998.90 |     10 | 2024-01-15  | Developer   | Mumbai |
|   105 | Vijay | 49999.00 |     20 | 2023-11-12  | Salesperson | Mumbai |
|   102 | Amit  | 55000.00 |     20 | 2023-06-10  | Manager     | Pune   |
|   103 | Priya | 66000.00 |     10 | 2022-03-20  | Salesperson | Mumbai |
+-------+-------+----------+--------+-------------+-------------+--------+
6 rows in set (0.00 sec)

mysql> ALTER TABLE Department
    -> RENAME COLUMN DeptName to DepartmentName;
Query OK, 0 rows affected (0.02 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> SELECT * FROM Department;
+--------+----------------+----------+
| DeptID | DepartmentName | Salary   |
+--------+----------------+----------+
|     10 | IT             | 50000.00 |
|     20 | Sales          | 45000.00 |
|     30 | HR             | 40000.00 |
|     40 | Finance        | 35000.00 |
|     50 | Marketing      | 42000.00 |
+--------+----------------+----------+
5 rows in set (0.00 sec)
```